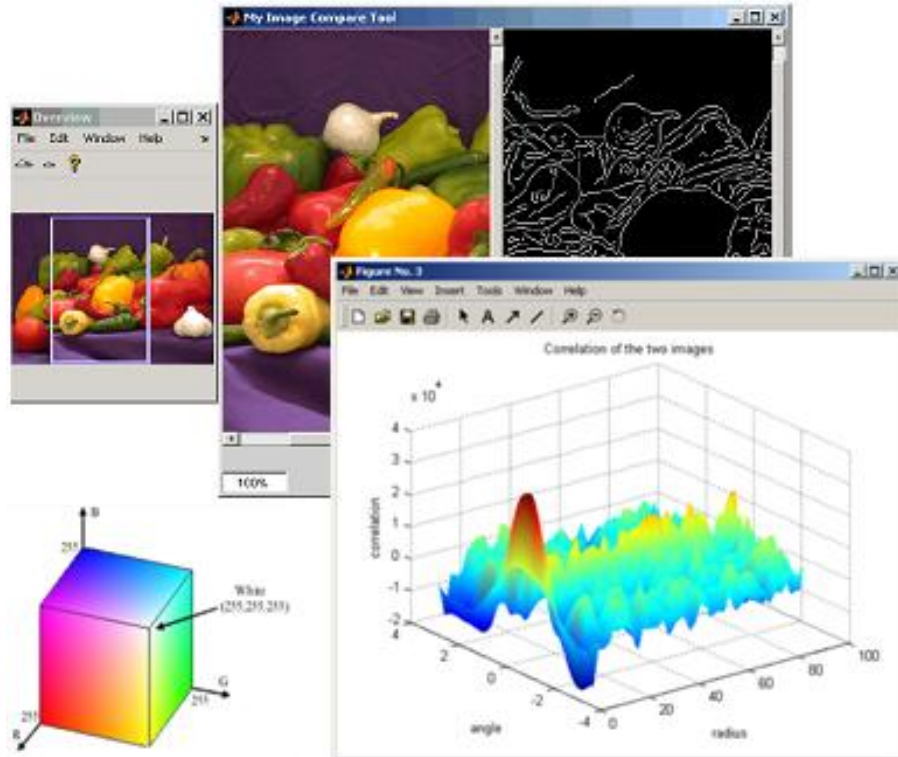


Digital Image Processing

LECTURE -4: DIGITAL IMAGE RESOLUTION AND ASPECT RATIO



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Assistant Professor
ECE Department
GLA University Mathura

Image Resolution

Digital Camera Resolution

SAMSUNG

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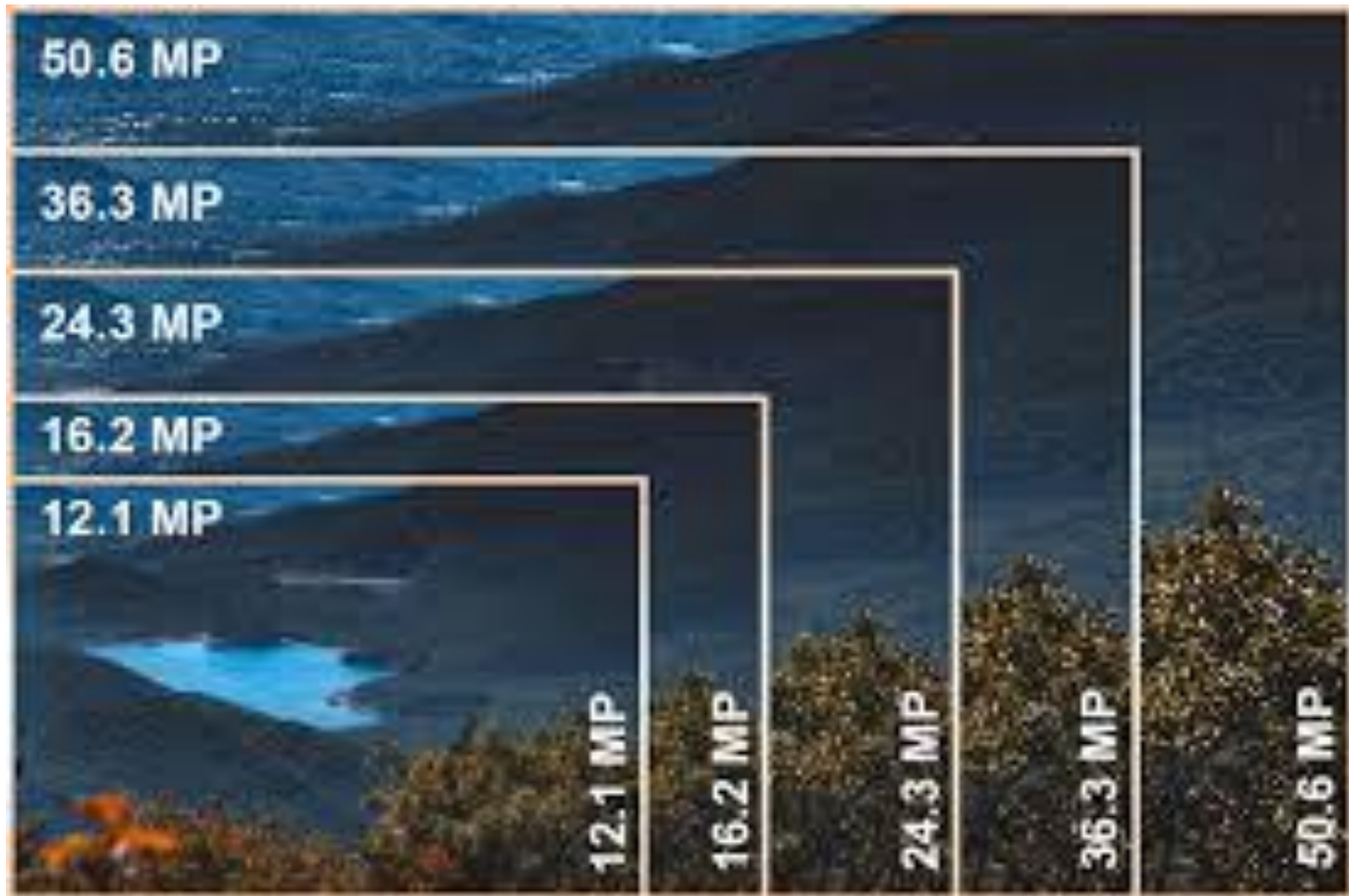
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Digital Camera Resolution



Digital Camera Resolution

SAMSUNG

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- Digital image resolution is a crucial aspect of digital photography and imaging, impacting visual quality across various applications.
- At its core, resolution refers to the level of detail that an image holds, which is significantly influenced by the number of pixels contained within the image.
- Higher resolution typically translates to increased detail and clarity, allowing for brighter and more vivid images.

Digital Camera Resolution

❑ Common Resolutions:

- **Low:** 640x480, 1024x768 (older devices, small screens)
- **Medium:** 1280x720 (HD), 1920x1080 (Full HD)
- **High:** 2560x1440 (QHD), 3840x2160 (4K), 7680x4320 (8K)

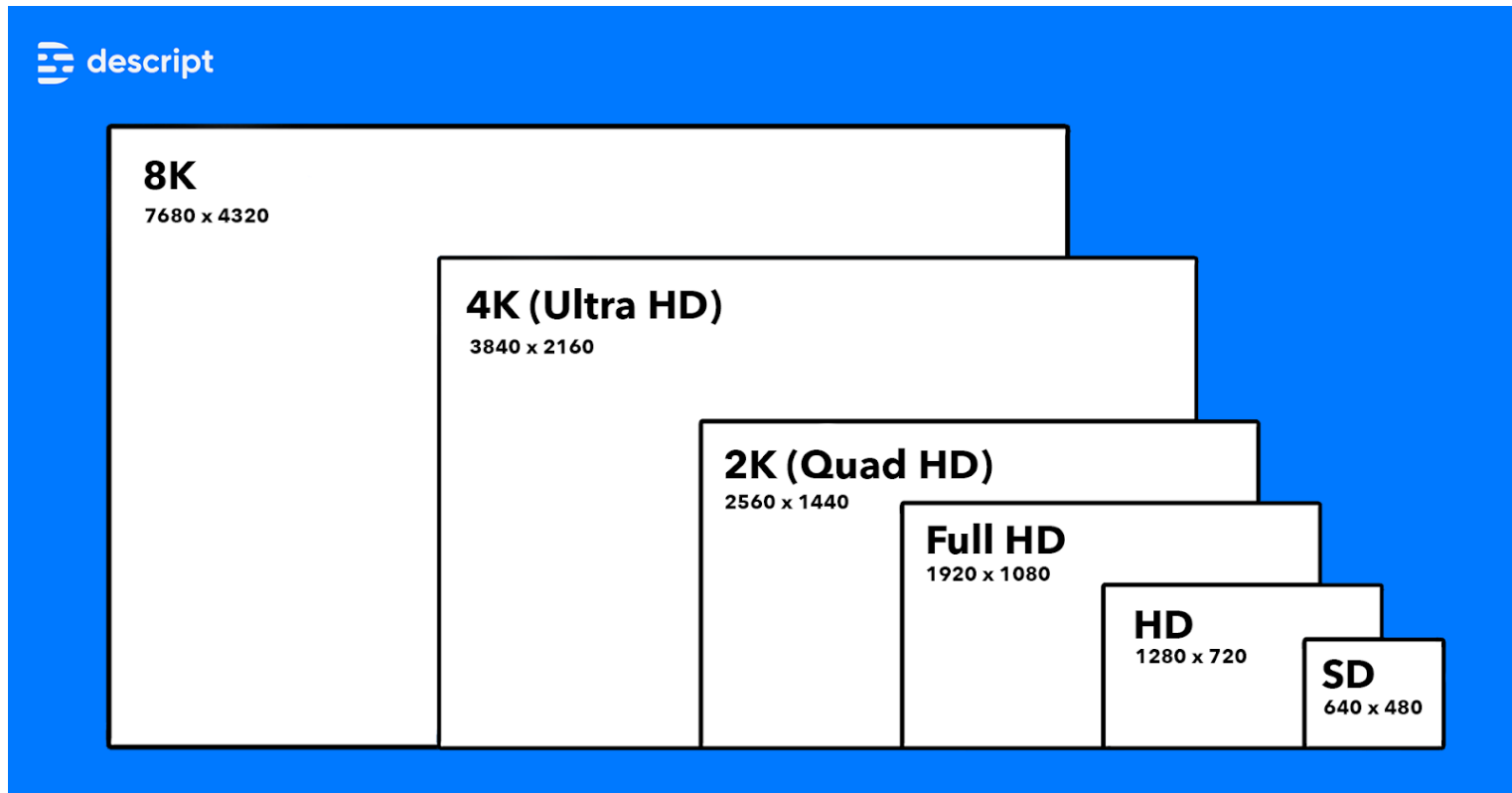


Image Resolution

❑ Resolution:

- It is the ability of the imaging system to produce the **smallest discernable details**,
- The **smallest sized object clearly and differentiate it from the neighbouring small objects** that are present in the image.

❑ Types of Resolution:

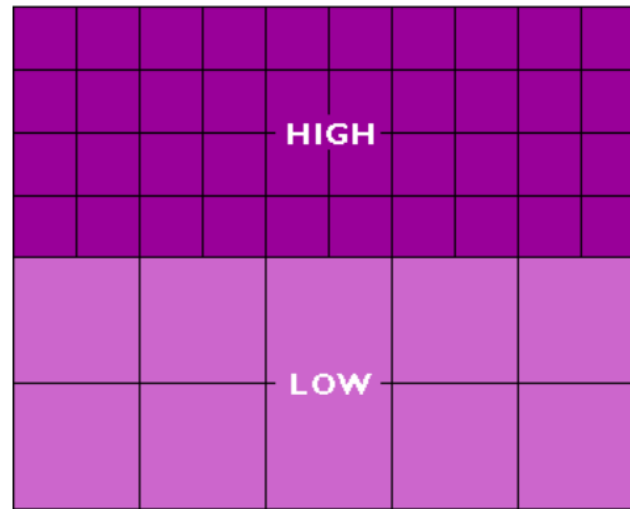
- Spatial resolution
- Gray label resolution
- Spectral resolution
- Temporal resolution
- Radiometric resolution

Spatial Resolution

❑ Spatial Resolution:

- The smallest possible feature that can be detected in an image
OR
- The size of single pixel or area covered by a single pixel (GSD)
- Spatial resolution depends on two parameters:
 - a) The number of pixels of the image
 - b) The number of bits necessary to adequate intensity resolution referred to as bit depth.
- Line pairs per unit distance
- Dots/pixels per unit distance (dots per inch – dpi)

Spatial Resolution



MODIS (250 m)



LANDSAT 8 (30 m)



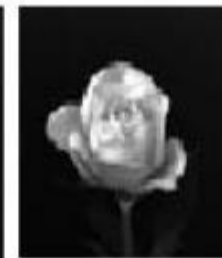
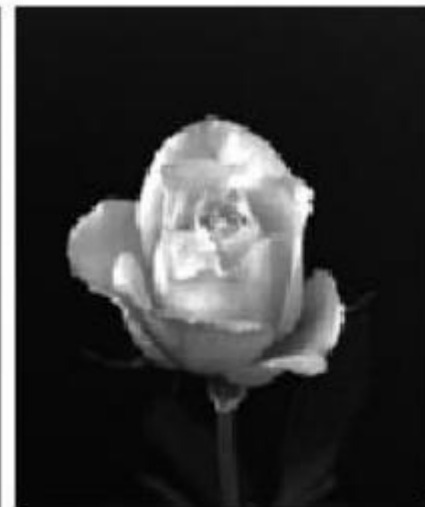
SENTINEL 2 (10 m)



DRONE (0.05 m)

Checkerboard Effect

A1024*1024, 8-bit image subsampled down to size 32*32 pixels. The number of allowable gray levels was kept at 256



32

64

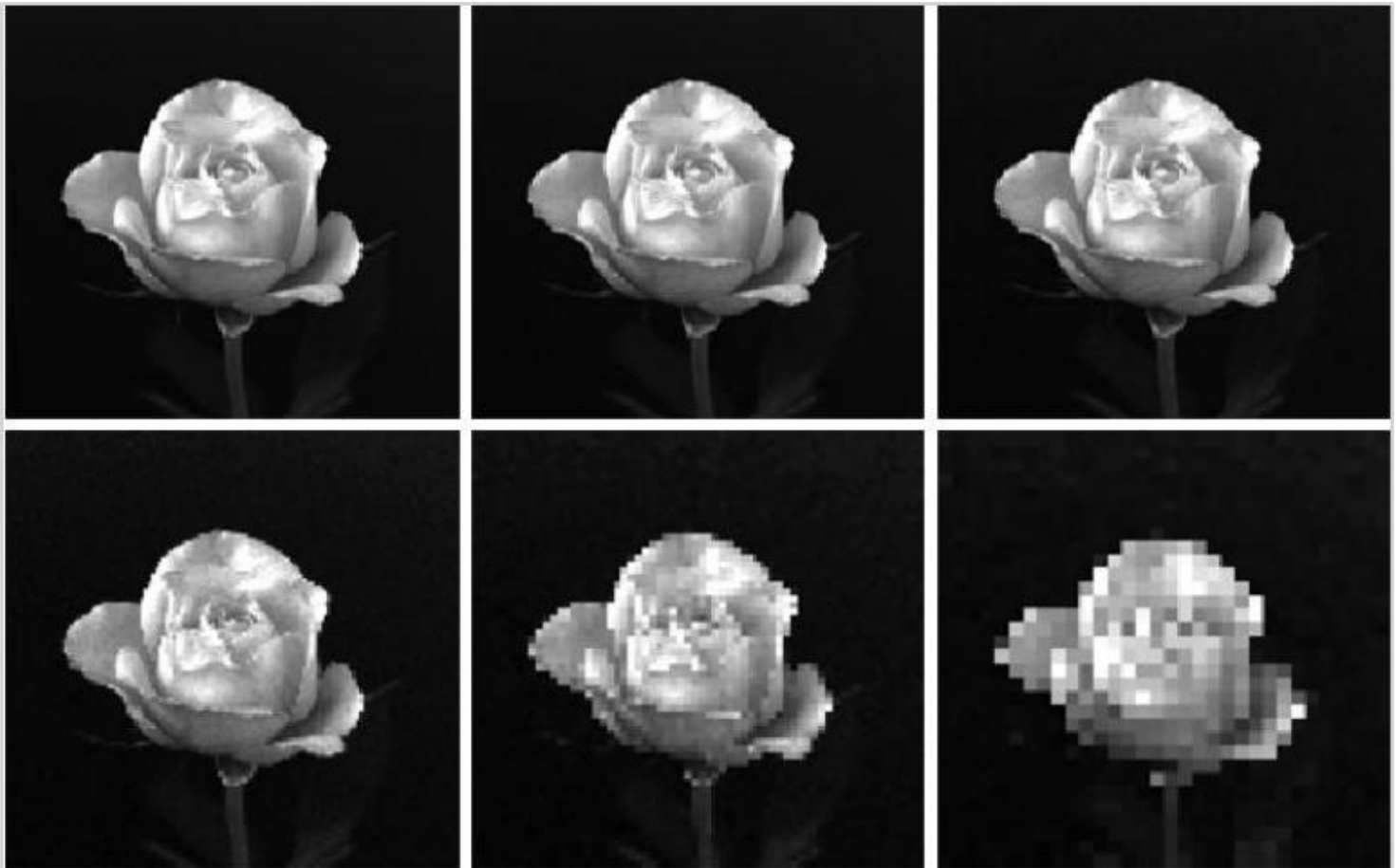
128

256

512

1024

Checkerboard Effect



(a) 1024×1024 , 8-bit image. (b) 512×512 image resampled into 1024×1024 pixels by row and column duplication. (c) through (f) 256×256 , 128×128 , 64×64 , and 32×32 images resampled into 1024×1024 pixels

Checkerboard Effect

When the no. of pixels in an image is reduced keeping the no. of gray levels in the image constant, fine **checkerboard patterns** are found at the edges of the image. **This effect is called the checker-board effect.**

Gray Scale Resolution

☐ **Grey level (Intensity) Resolution:** Gray-level resolution similarly refers to the smallest detectable change in gray level.

Original Image A



Image with 128 Gray Shades

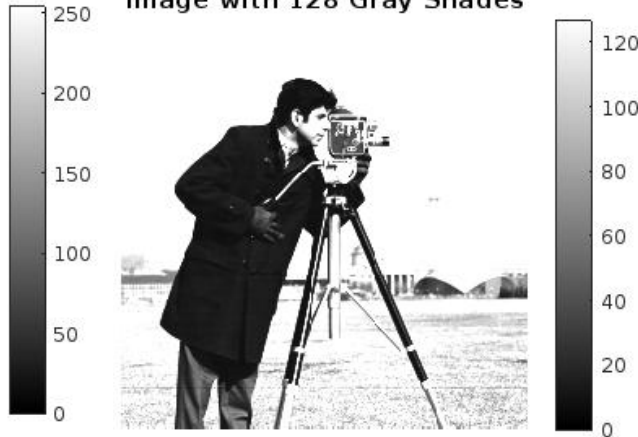


Image A with 16 Gray Shades

Image with 64 Gray Shades



Image with 32 Gray Shades

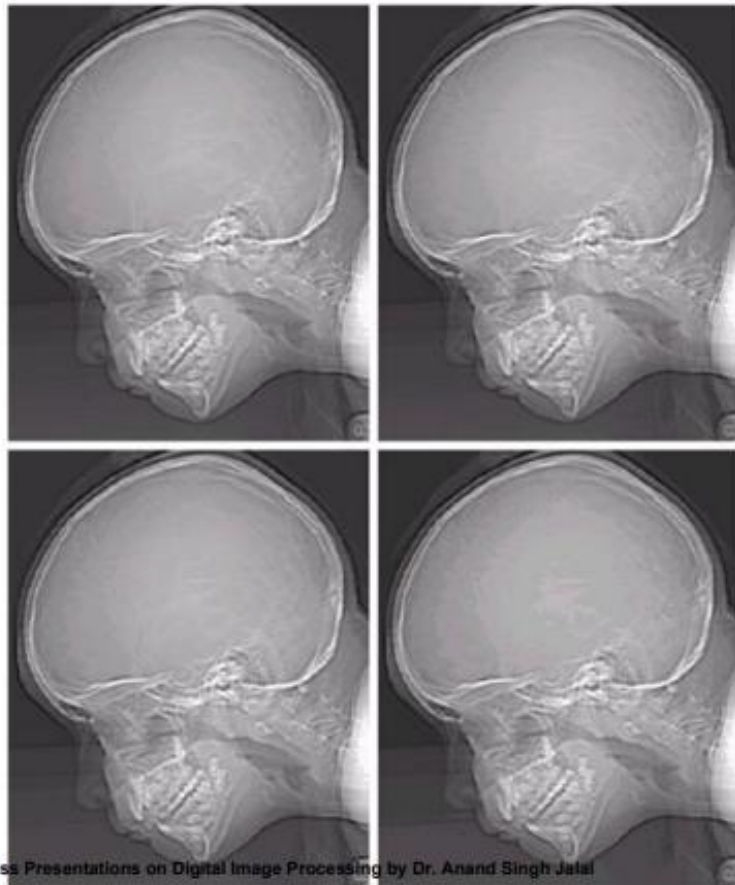


False Contouring

When the no. of gray-levels in the image is low, the **foreground details of the image merge with the background details** of the image, causing ridge like structures. **This degradation phenomenon is known as false contouring.**

Gray Scale Resolution

☐ **Grey level (Intensity) Resolution:** Gray-level resolution similarly refers to the smallest detectable change in gray level.



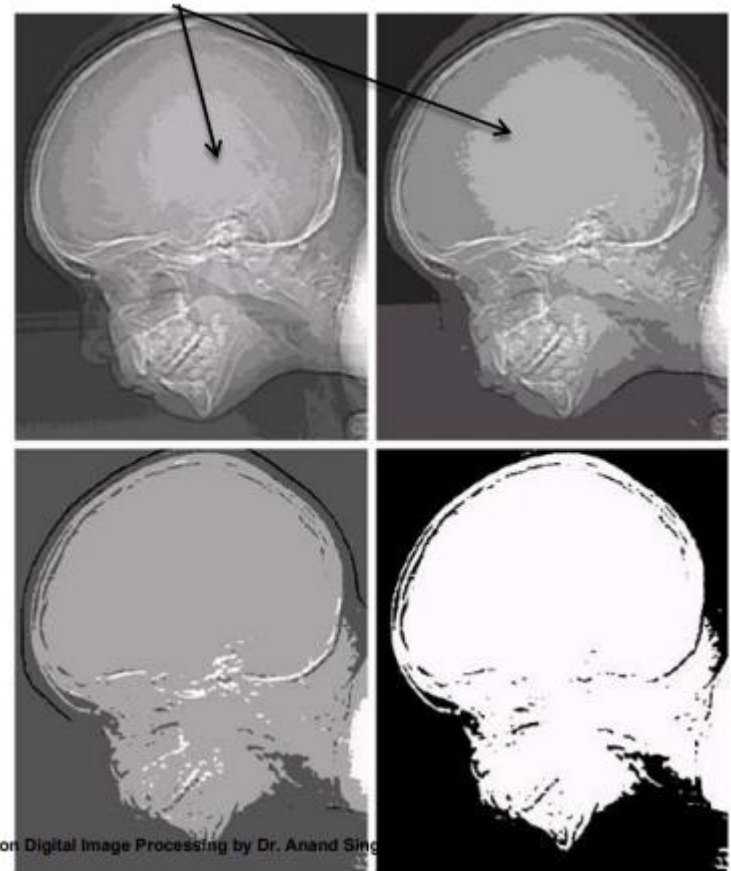
a b
c d

(a) 452×374 , 256-level image. (b)–(d) Image displayed in 128, 64, and 32 gray levels, while keeping the spatial resolution constant.

e f
g h

(Continued)
(e)–(h) Image displayed in 16, 8, 4, and 2 gray levels. (Original courtesy of Dr. David R. Pickens, Department of Radiology & Radiological Sciences, Vanderbilt University Medical Center.)

False contouring



Resolution: How Much Is Enough?

The big question with resolution is always how much is enough?

- This all depends on what is in the image and what you would like to do with it
- Key questions include
 - Does the image look pleasing?
 - Can you see what you need to see within the image?

Resolution: How Much Is Enough?



The picture on the right is fine for counting the number of cars, but not for reading the number plate

University Questions

Ques 1: One mobile phone has higher spatial resolution and another has higher Gray level resolution. Which mobile would you prefer to buy.

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Ques 2: Define Image Resolution. What are the spatial resolution and intensity resolution. What kind of resolution do you prefer in your mobile phone screen.

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University Questions

Ques 3: What is Checkerboard Effect?

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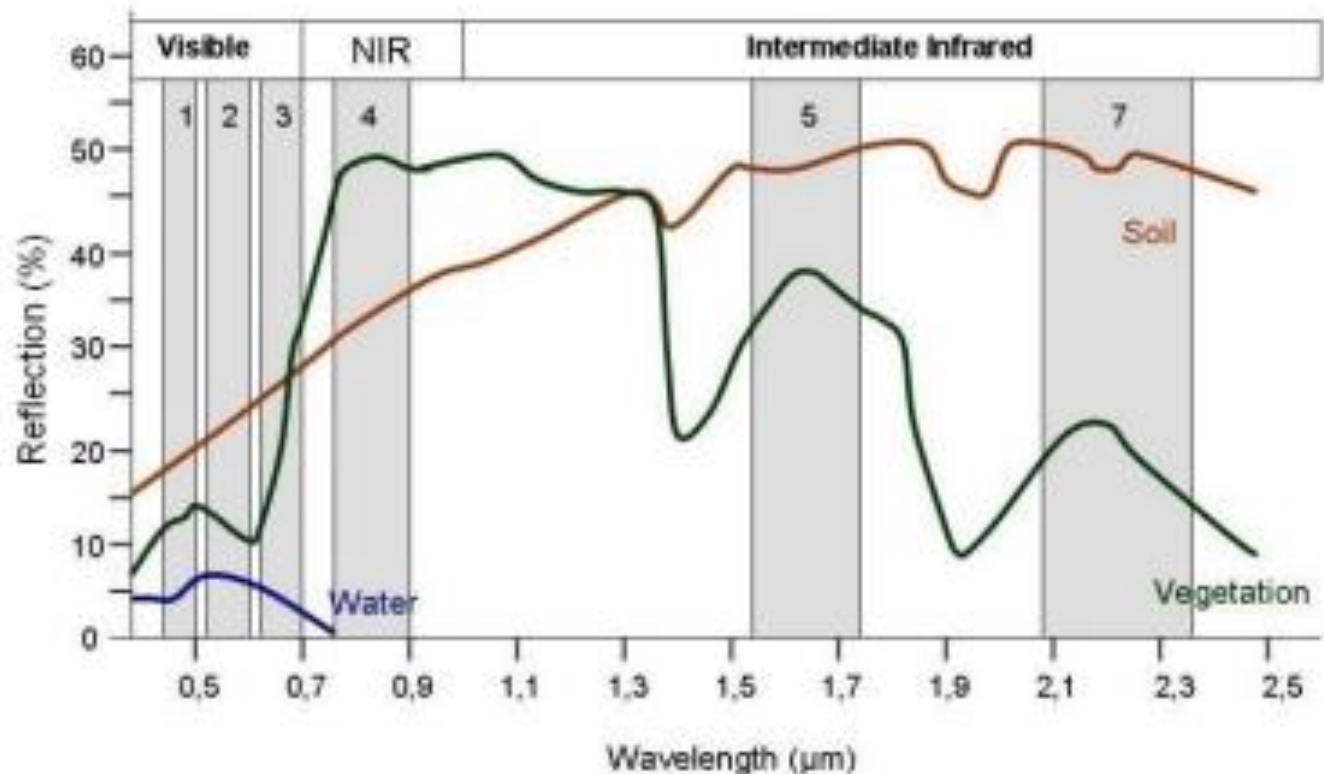
Ques 4: Differentiate between Spatial and Gray Level resolution in brief.

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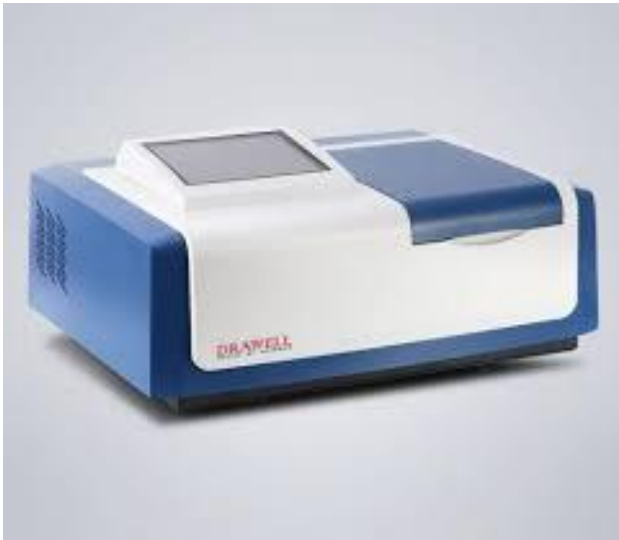
Spectral Resolution

□ Spectral Resolution

- It describes the capability of a sensor to capture information in various bands which have fine wavelength intervals
- Information recorded by the sensor in various bands are referred as multi-spectral sensors



Spectral Resolution



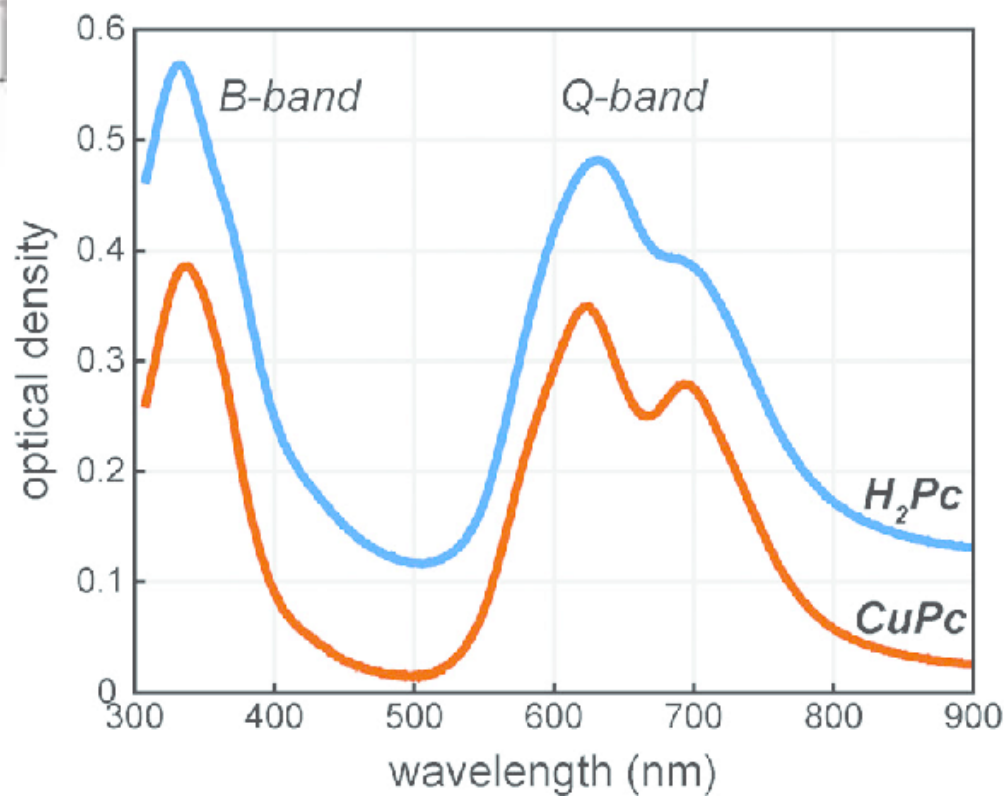
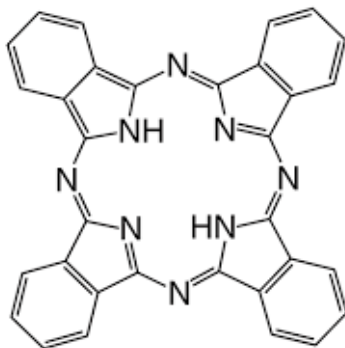
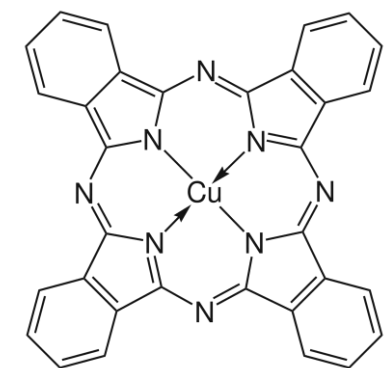
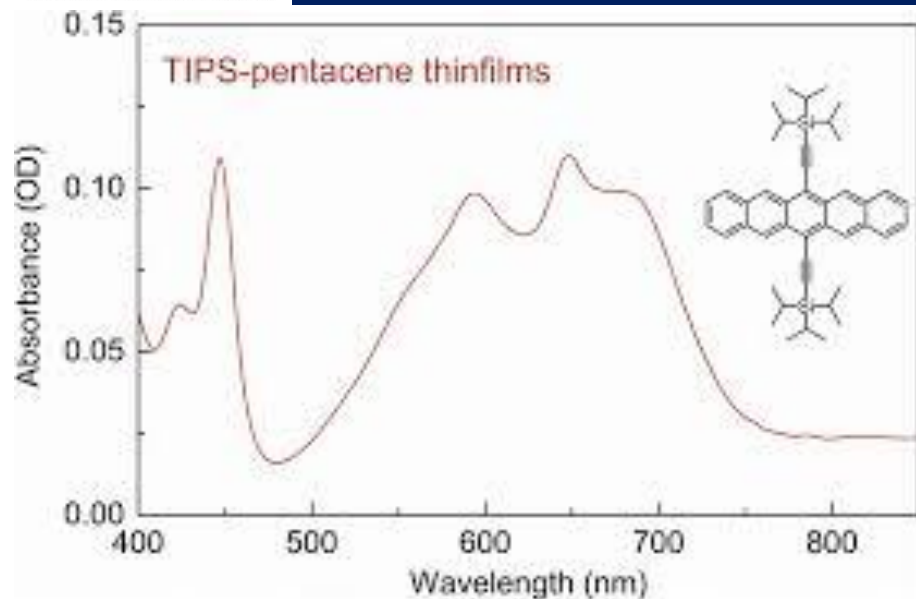
UV-Vis Spectroscopy

FTIR Spectroscopy



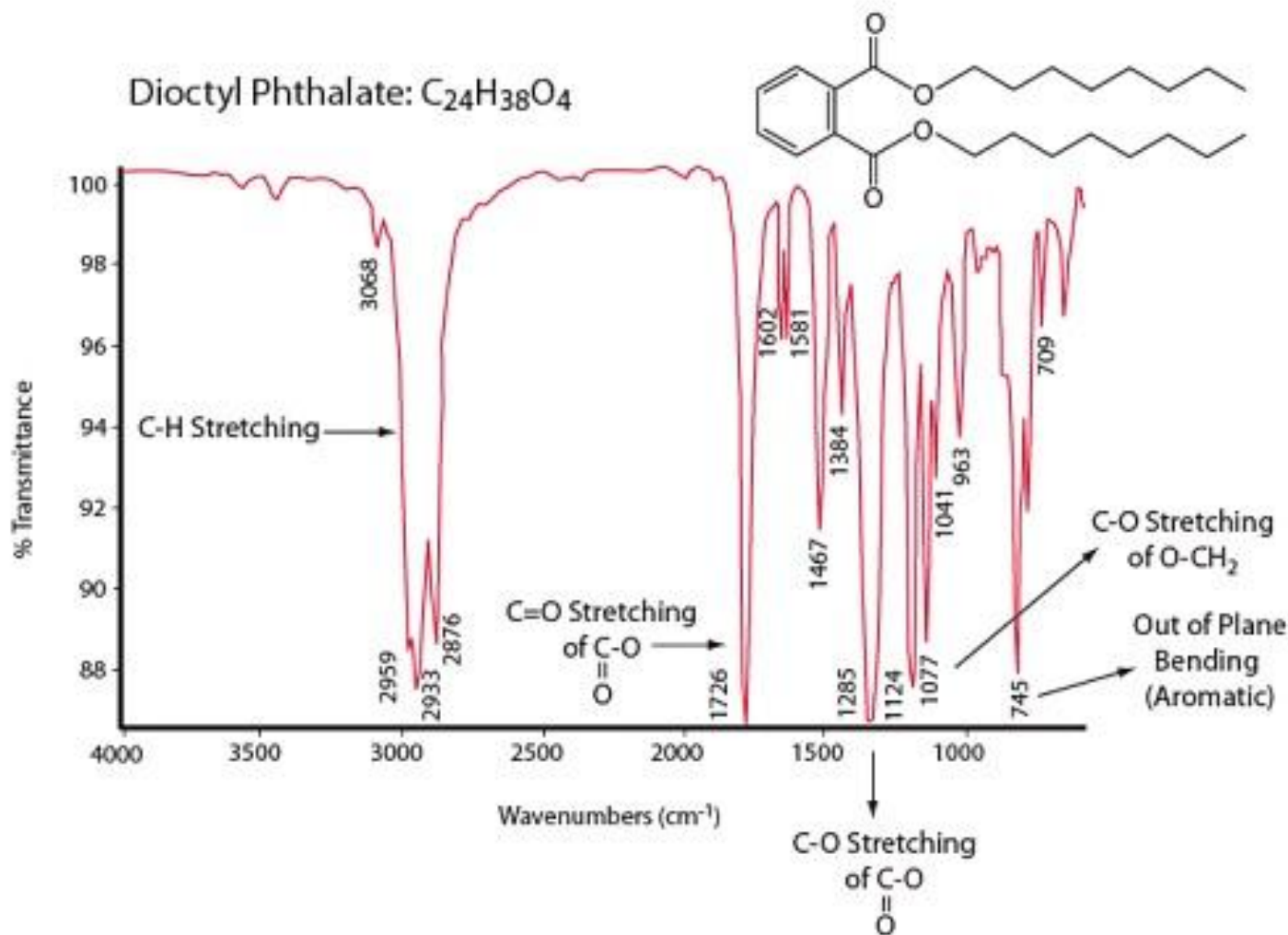
Spectral Resolution

UV-Vis Spectroscopy



Spectral Resolution

FTIR Spectroscopy

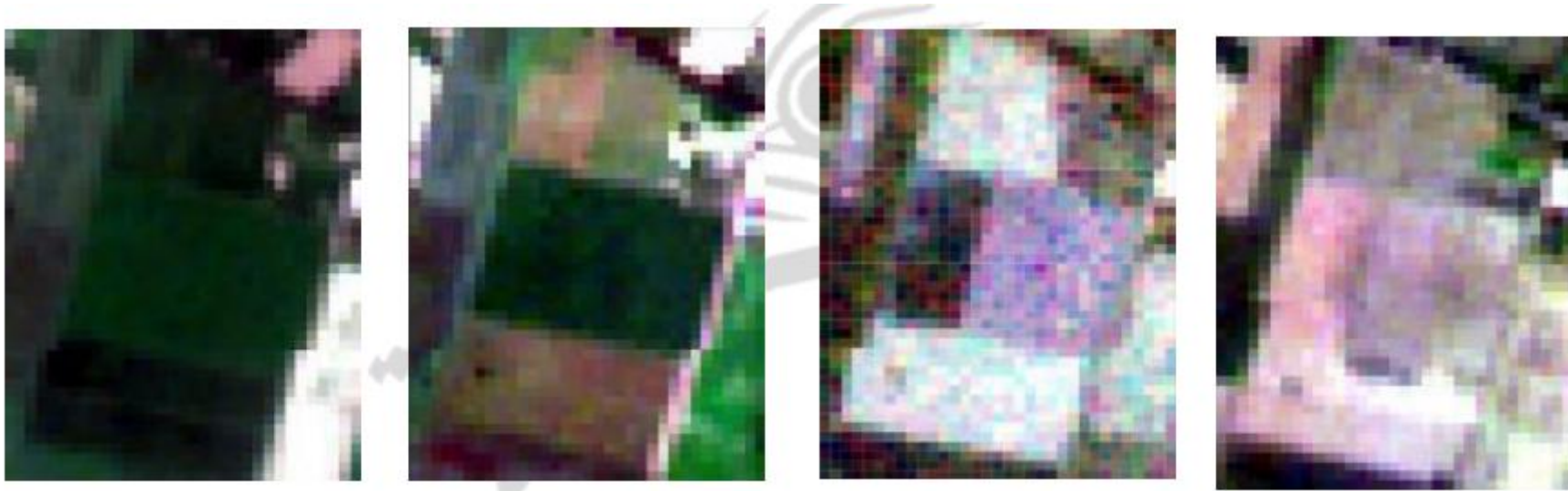


Temporal Resolution

□ Temporal Resolution:

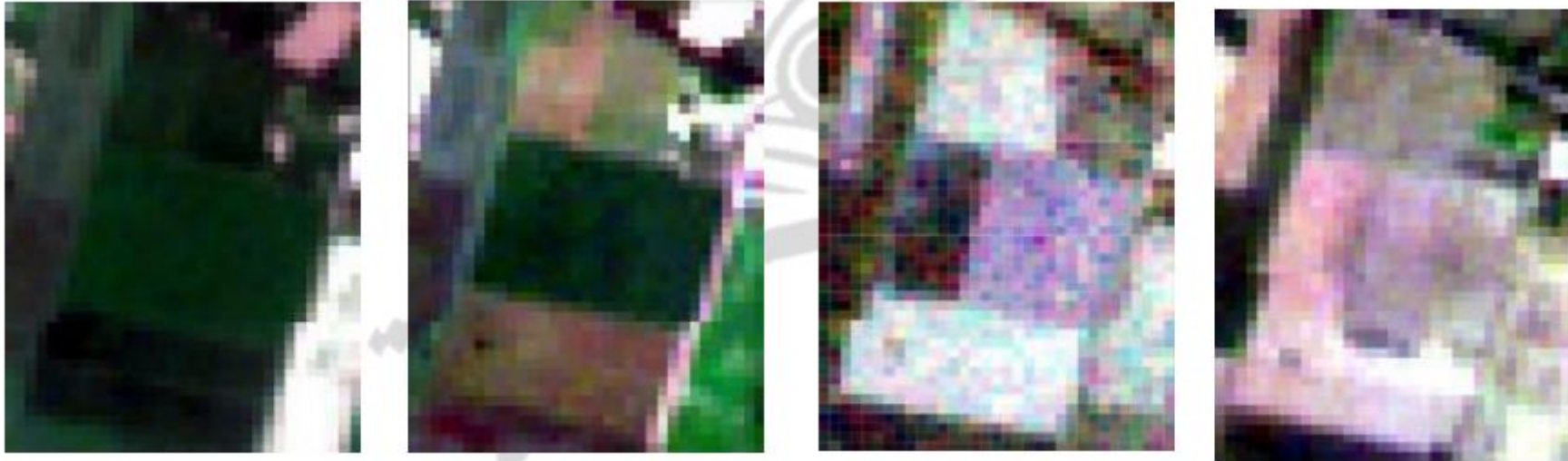
- The concept of revisit period is important for monitoring that help us to **identify the changes over a period of time**
- Time taken by a sensor to locate exact same area second time at same interval time is considered as temporal resolution

Sentinel-2
RGB

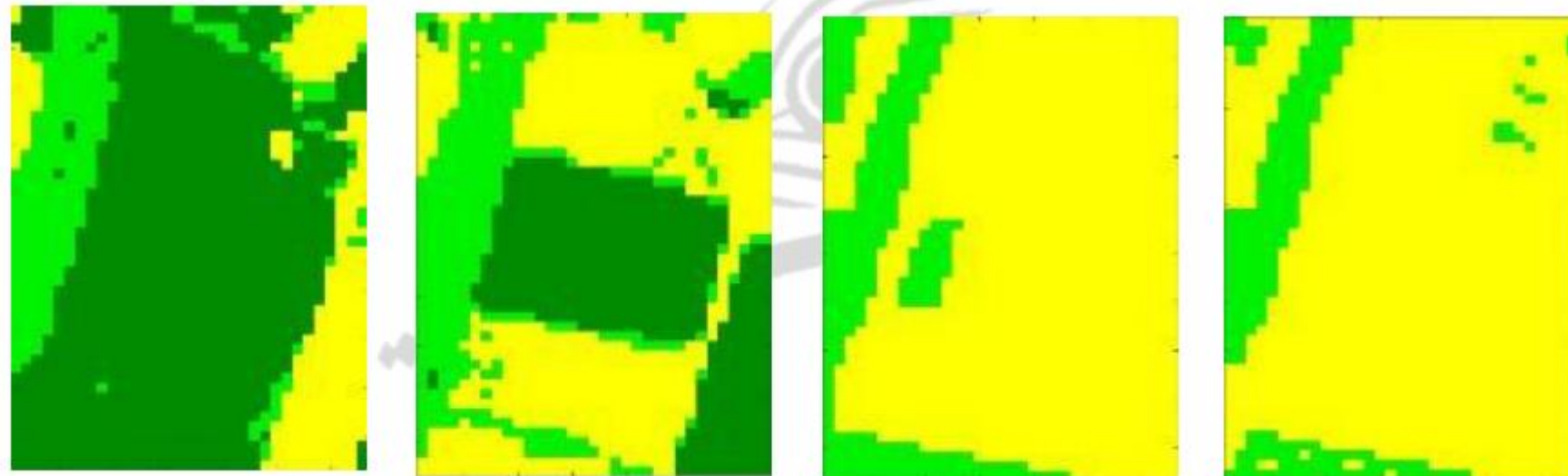


Temporal Resolution

Sentinel-2
RGB



Sentinel-2
Classified



Bare Land

Dense Vegetation

Sparse Vegetation

Radiometric Resolution

- Radiometric resolution is a critical aspect of remotely sensed data that directly affects the quality and accuracy of the images produced by satellite sensors.
- It refers to the ability of a sensor to **detect and distinguish between different levels of reflected or emitted radiation from the Earth's surface**, essentially measuring the sensitivity of the sensor to variations in radiance.
- Radiometric resolution is a measure of how well a sensor can differentiate between small differences in energy.



University Questions

Ques 5: Suggest the suitable resolution for the following scenario:

- A user would like to monitor the growth of wheat crop from the sowing time till it gets harvested.
- A tropical cyclone arises in the southern part of the Bay of Bengal and is moving upwards which is tracked by a user.
- A user would like to have a sensor which he is not bothering about the pixel size but he wants more number of shades in the image

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Image Aspect Ratio

Image Aspect Ratio

- An aspect ratio is a proportional relationship between an image's width and height

$$\text{Image Aspect Ratio} = \frac{\text{Width}}{\text{Height}}$$

Image Aspect Ratio

WIDTH

HEIGHT



Image Aspect Ratio



Width = 12, Height = 6.77



Width = 6.77, Height = 6.77



Width = 6.77, Height = 12

Image Aspect Ratio



Image Aspect Ratio

- ❑ **1:1 Image Aspect Ratio:** encourages a different kind of focus and composition, often used for social media platforms like Instagram

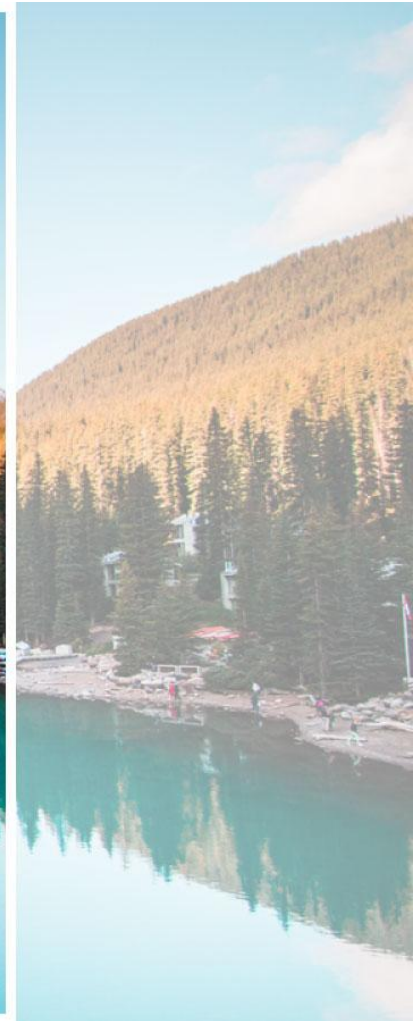
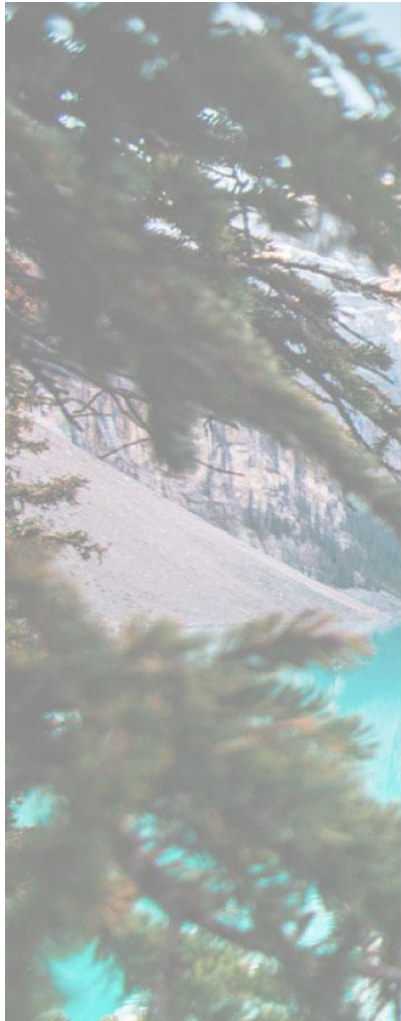


Image Aspect Ratio

- ❑ **3:2 Image Aspect Ratio:** used for prints and digital images (DSLR and Tablets) since it correlates well with popular print sizes.

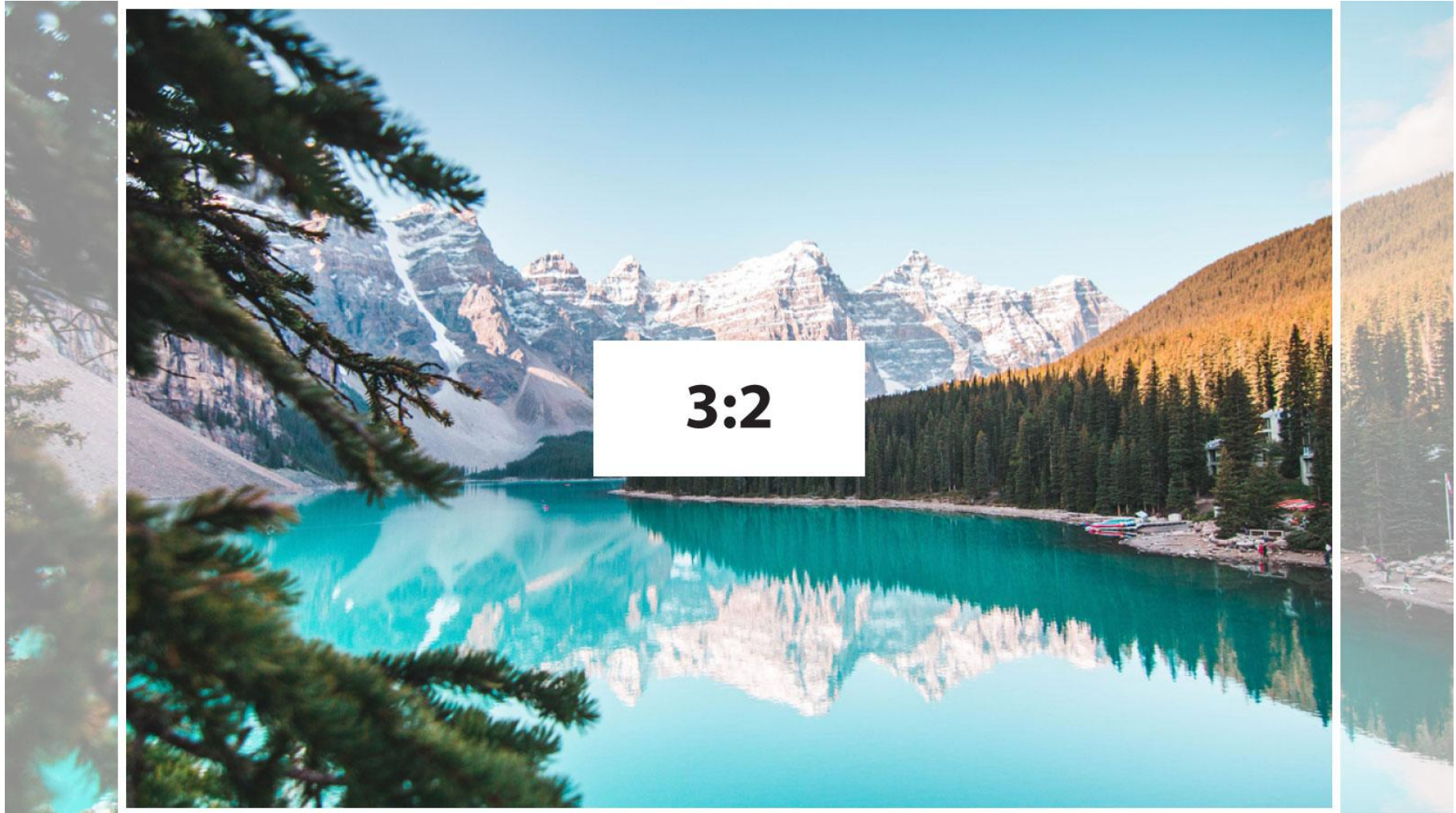


Image Aspect Ratio

❑ **4:3 Image Aspect Ratio** : also known as [the academy ratio](#), A classic ratio used in older televisions and some digital cameras.

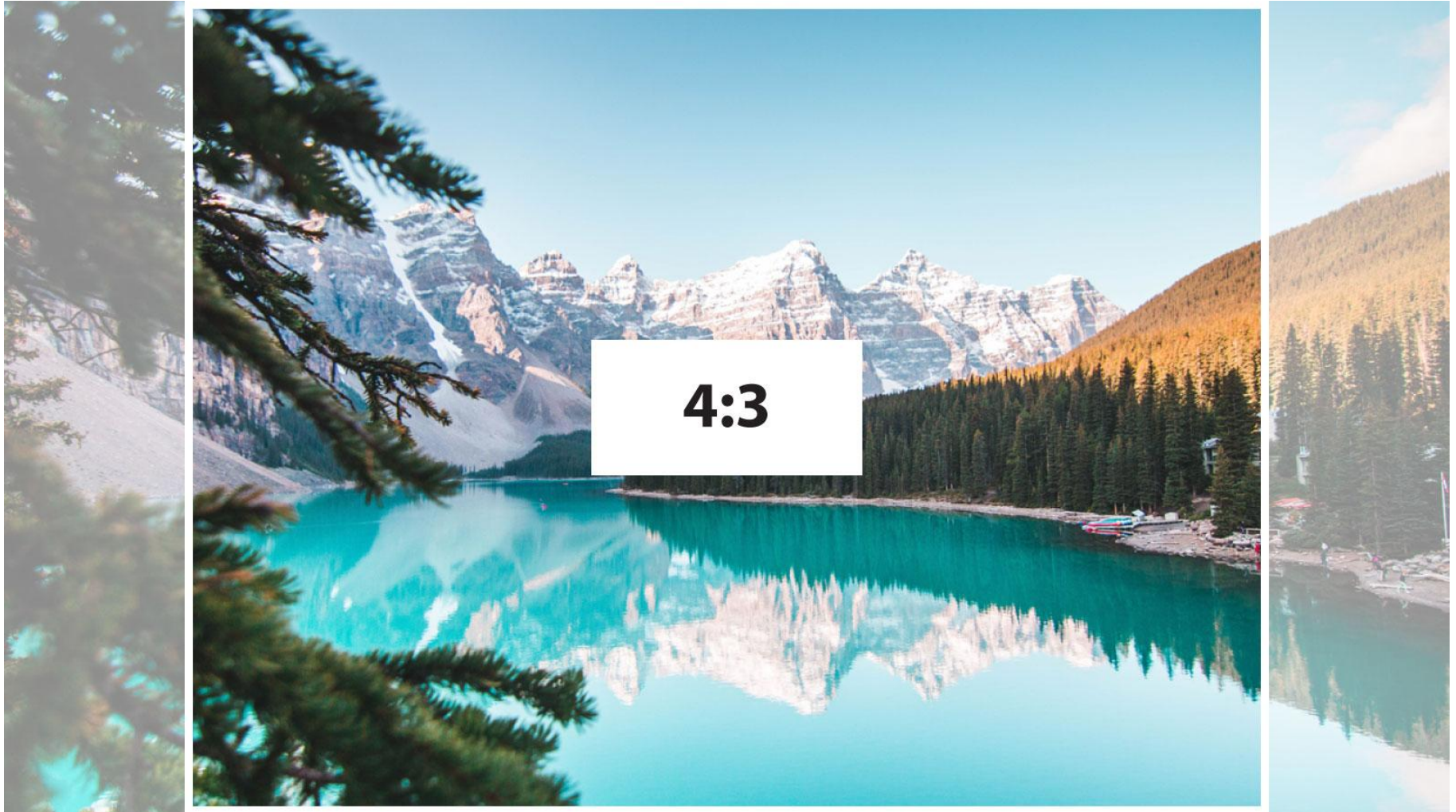


Image Aspect Ratio

❑ **9:16 Image Aspect Ratio:** used in smartphones and social platforms like TikTok, Snapchat, Instagram Reels and YouTube Shorts.

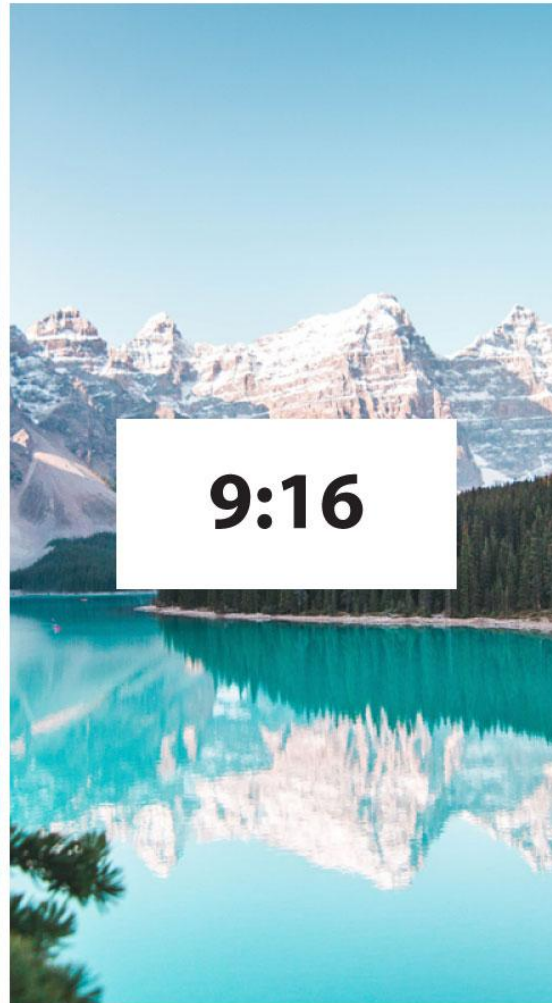


Image Aspect Ratio

- ❑ **16:9 Image Aspect Ratio** : The standard for widescreen televisions, movies, online video platforms and many modern displays.



Image Aspect Ratio

- ❑ **21:9 Image Aspect Ratio:** The 21:9 ratio is the best aspect ratio for cinematic videos.



Numerical Examples

Ques1: If we want to resize a 1024x768 image to one that is 600 pixels wide with the same aspect ratio as the original image, what should be the height of the resized image?

Numerical Examples

Ques2: Suppose, we have an image that have 16 rows and have a aspect ratio of 1 or 1:1. Find the total number of pixels in the image.

Numerical Examples

Ques3: Suppose, an image have an aspect ratio of 3:2 with a width of 36 mm. Find the height of the image.

Numerical Examples

Ques4: Suppose we have an image that have 20 rows and have an aspect ratio of 2:5. Find the total number of pixel in an image.

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Numerical Examples

Ques 5: Suppose there is an Image I1 having the 20 rows and 10 columns. There is another image I2 having 100 rows and have the same aspect ratio as of I1. In Image I2, the maximum value among all the pixels is not exceeding 16. The Image I2 need to be transferred from one place to the other that have a transfer speed of 20 bits/sec. The image is transferred in the form of packets having the packet size of 10 bits and for segregating the packet they are using the flag bits at the beginning and at the end of the each packet such that the bits used in the flags are $\frac{1}{2}$ of the packet size. Compute the transfer time in sec.

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Thank You